

GPPU Experiment P1 (2025)

2026/2/24 – 2/26

RCNS I-03

Name:

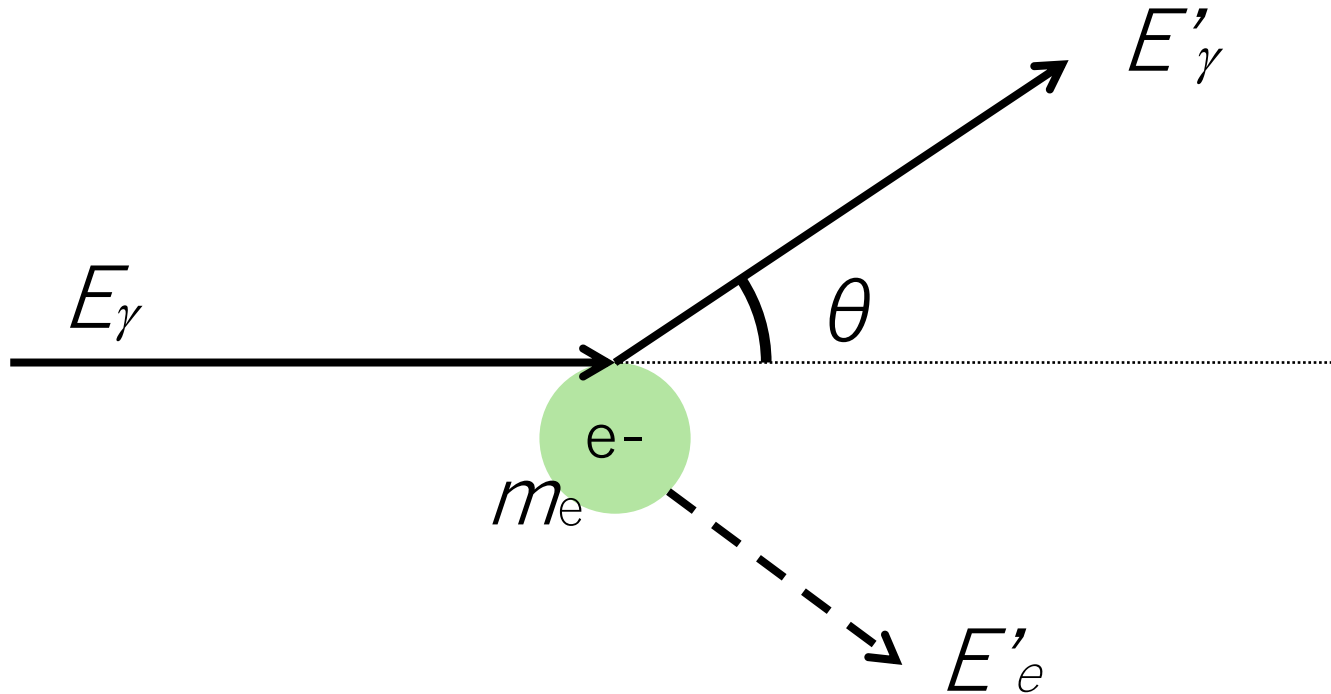
E-mail:

Finish date/time : 2026/2/ :

Rule

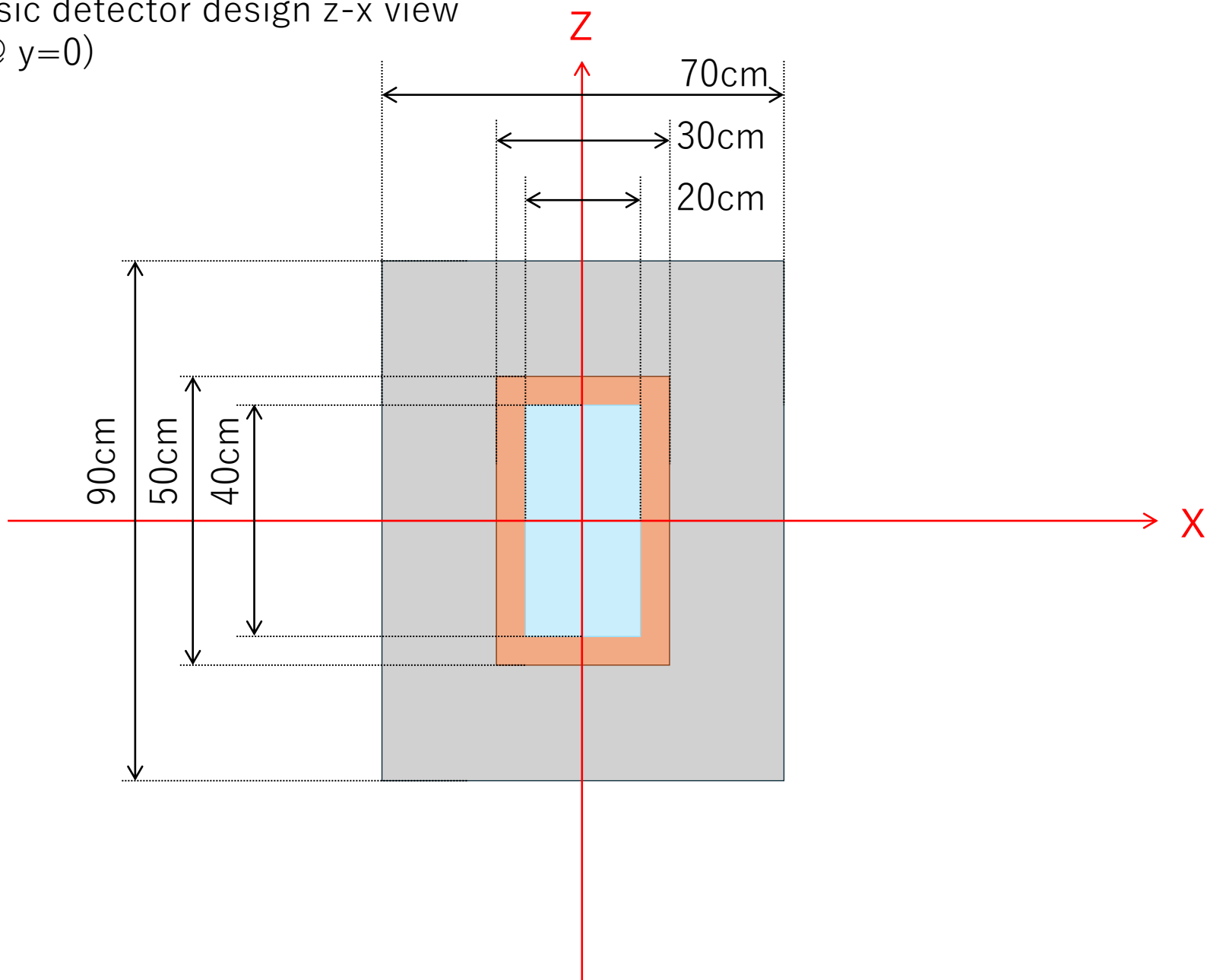
- Do not rewrite program codes used in your laboratory or written by your senior students.
- Please write your code by yourself or using the base codes in the laptop PC.

1.1 Considering Compton scattering, show E'_γ and E'_e using E_γ , m_e and θ .

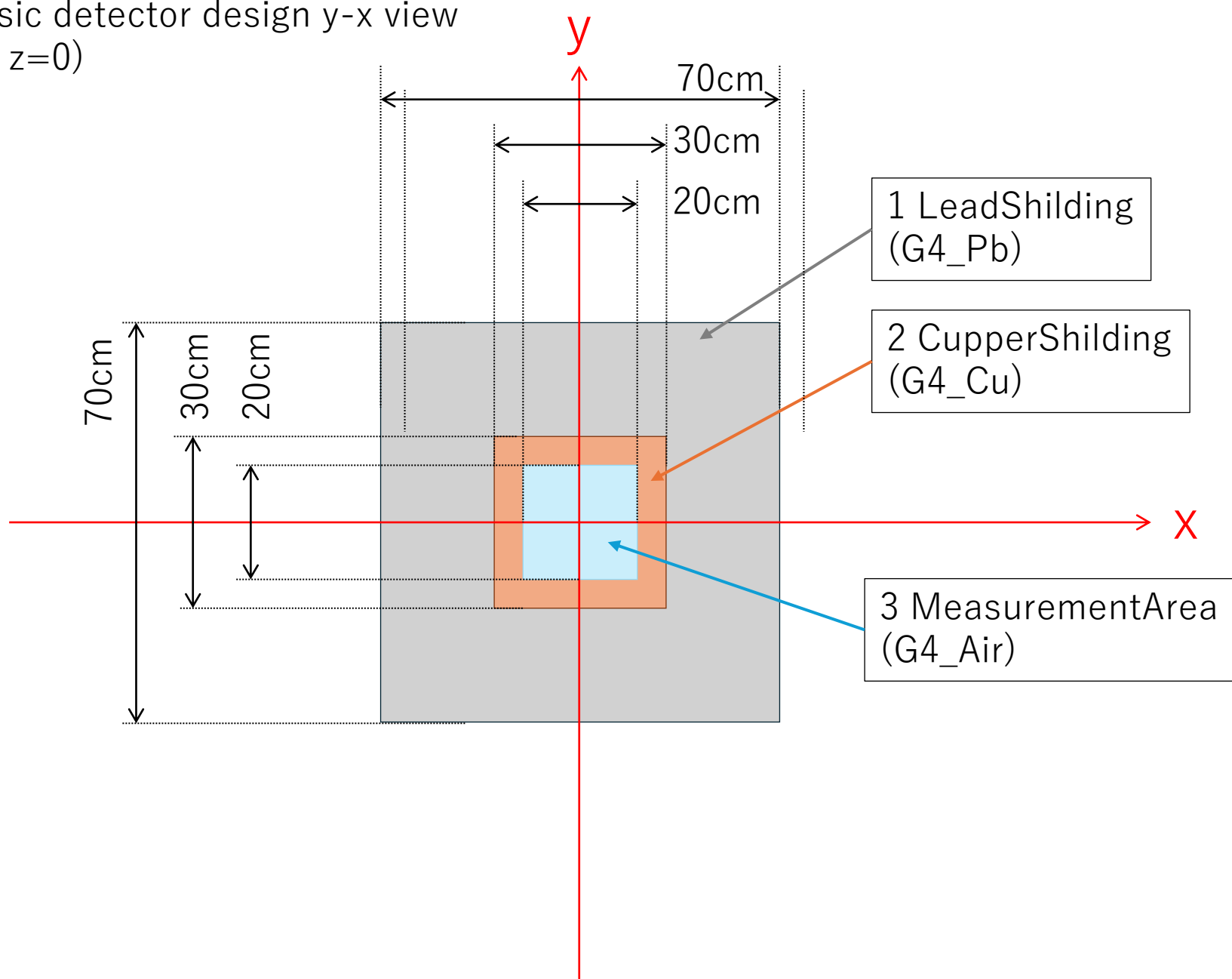


1.2 Calculate E'_γ and E'_e when $E_\gamma=1.0[\text{MeV}]$, $m_e=0.511$ $[\text{MeV}/c^2]$ and $\theta=180^\circ$.

2.1 Basic detector design z-x view (@ $y=0$)

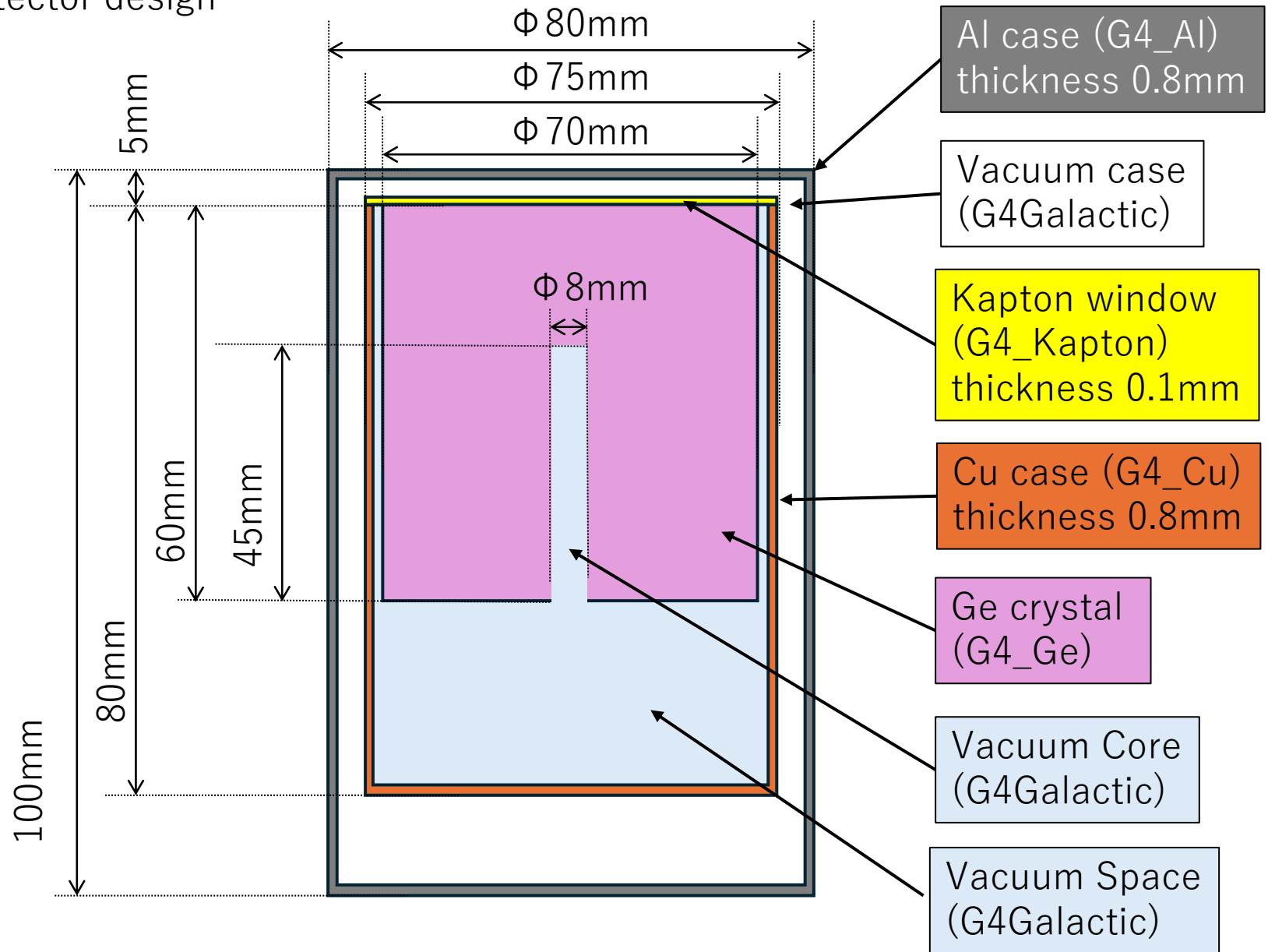


2.2 Basic detector design y-x view (@ $z=0$)



3 Ge detector design

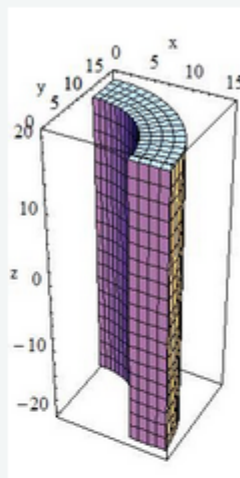
Z



Cylindrical Section or Tube:

Similarly to create a **cylindrical section** or **tube**, one would use the constructor:

```
G4Tubs(const G4String& pName,  
        G4double pRMin,  
        G4double pRMax,  
        G4double pDz,  
        G4double pSPhi,  
        G4double pDPhi)
```



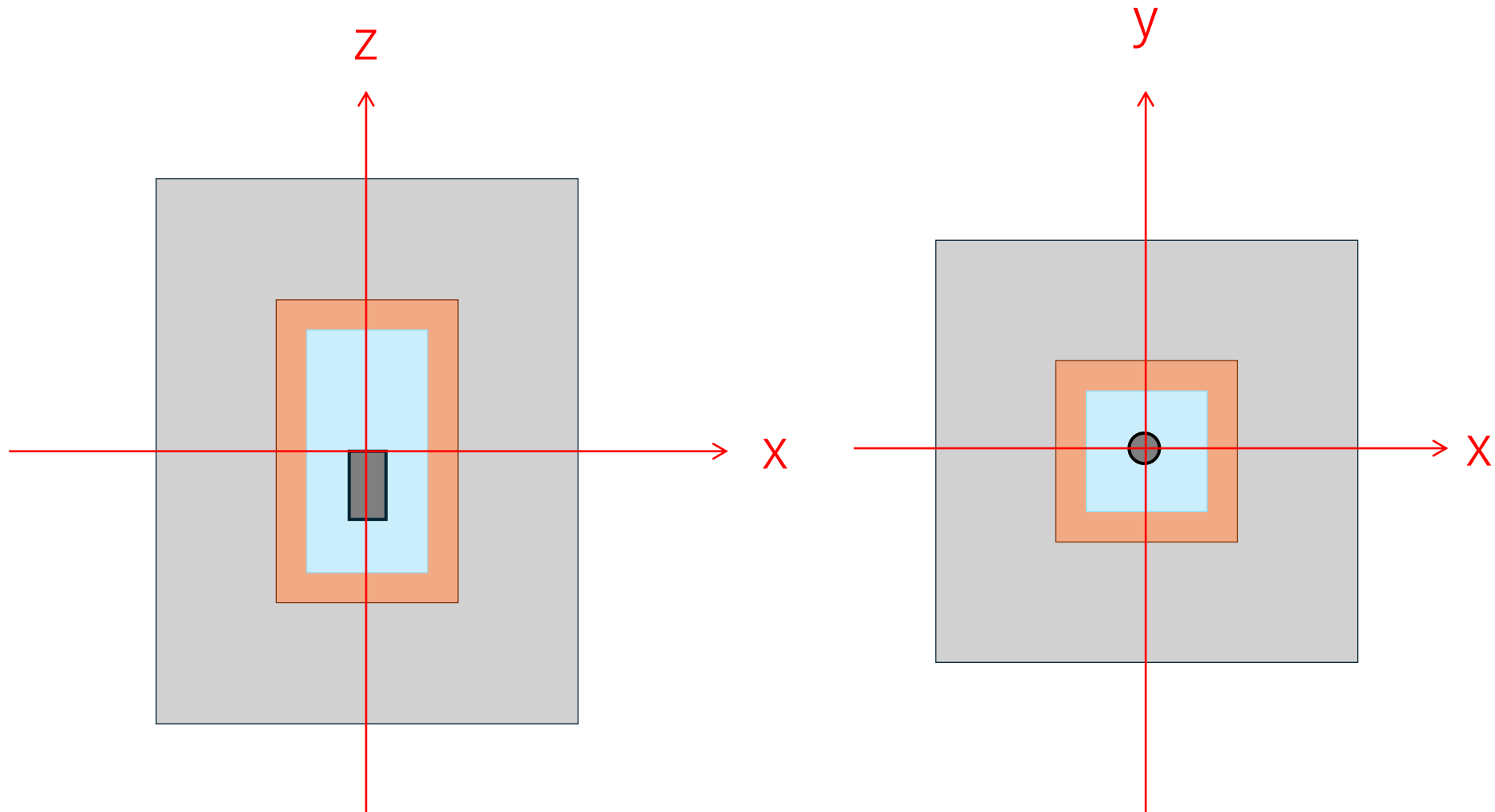
In the picture:

$pRMin = 10$, $pRMax = 15$, $pDz = 20$

giving its name $pName$ and its parameters which are:

$pRMin$	Inner radius	$pRMax$	Outer radius
pDz	Half length in z	$pSPhi$	Starting phi angle in radians
$pDPhi$	Angle of the segment in radians		

3.2 Ge detector Position



Set top-center part of Al case to (0,0,0)

4 Check Geometry using geantino

4.1 Check overlap

[Past your output]

[Past your screen capture of Qt viewer]

4.2 Show trajectories output of running geantino from position (0,0,490mm) and direction (0,0,-1). [Past your output /tracking/verbose 1]
Check crossing positions of each geometrical elements are same as your expectations.

[Past your output]

4.3 Show trajectories output of running geantino from position (0,10mm,490mm) and direction (0,0,-1).
Check crossing positions of each geometrical elements are same as your expectations.

[Past your output]

5.1 Activate Ge detector as a sensitive detector and record the energy deposition of each events and primary vertex information into root/csv file with tuple format .

Tuple format

- Name : Edep
- 0th column : Event number (integer)
- 1st column : Total energy deposit in Ge crystal (double)
- 2nd column : Primary vertex position x [mm] (double)
- 3rd column : Primary vertex position y [mm] (double)
- 4th column : Primary vertex position z [mm] (double)
- 5th column : Primary particle direction vector x (double)
- 6th column : Primary particle direction vector y (double)
- 7th column : Primary particle direction vector z (double)

5.2 Generate 1000000 event of 1.0 MeV gamma position(0,0,10mm) and direction (0,0,1) . Make a histogram of energy deposition in Ge crystal and calculate the efficiency of detecting full energy deposition (gamma events shown as 1.0 MeV).

[Past your macro file]

[Figure of Total energy deposit in Ge crystal]

Detection efficiency of 1 MeV gamma event: _____ $\pm$ _____ [stat.]

5.3 Generate 1000000 event of 1.0 MeV gamma position(0,0,10mm) and uniformed direction . Make a histogram of energy deposition in Ge crystal and calculate the efficiency of detecting full energy deposition (gamma events shown as 1.0 MeV). Make figures of primary particle directions (ϕ and $\cos\theta$ distribution)

[Past your macro file]

[Figure of Total energy deposit in Ge crystal]

Detection efficiency of 1 MeV gamma event: _____ $\pm$ _____ [stat.]

[ϕ distribution of primary particle direction vector figure]

[cos θ distribution of primary particle direction vector figure]

5.4 Do 5.2 as different gamma energies [0.25MeV, 0.5MeV, 0.75MeV, 1.25MeV, 1.5MeV,2.0MeV,2.5MeV]. Make histograms of each energies and calculate detection efficiency.

Make table and figure of detection efficiency as a function of gamma energy.

Gamma Energy	Detection Efficiency (w/ stat. error)
0.25	$\pm$
0.5	$\pm$
0.75	$\pm$
1.0	$\pm$
1.25	$\pm$
1.5	$\pm$
2.0	$\pm$
2.5	$\pm$

0.25MeV

0.5MeV

0.75MeV

1.25MeV

1.5MeV

2.0MeV

2.5MeV

Add comment for each peaks and edges shown in this figure

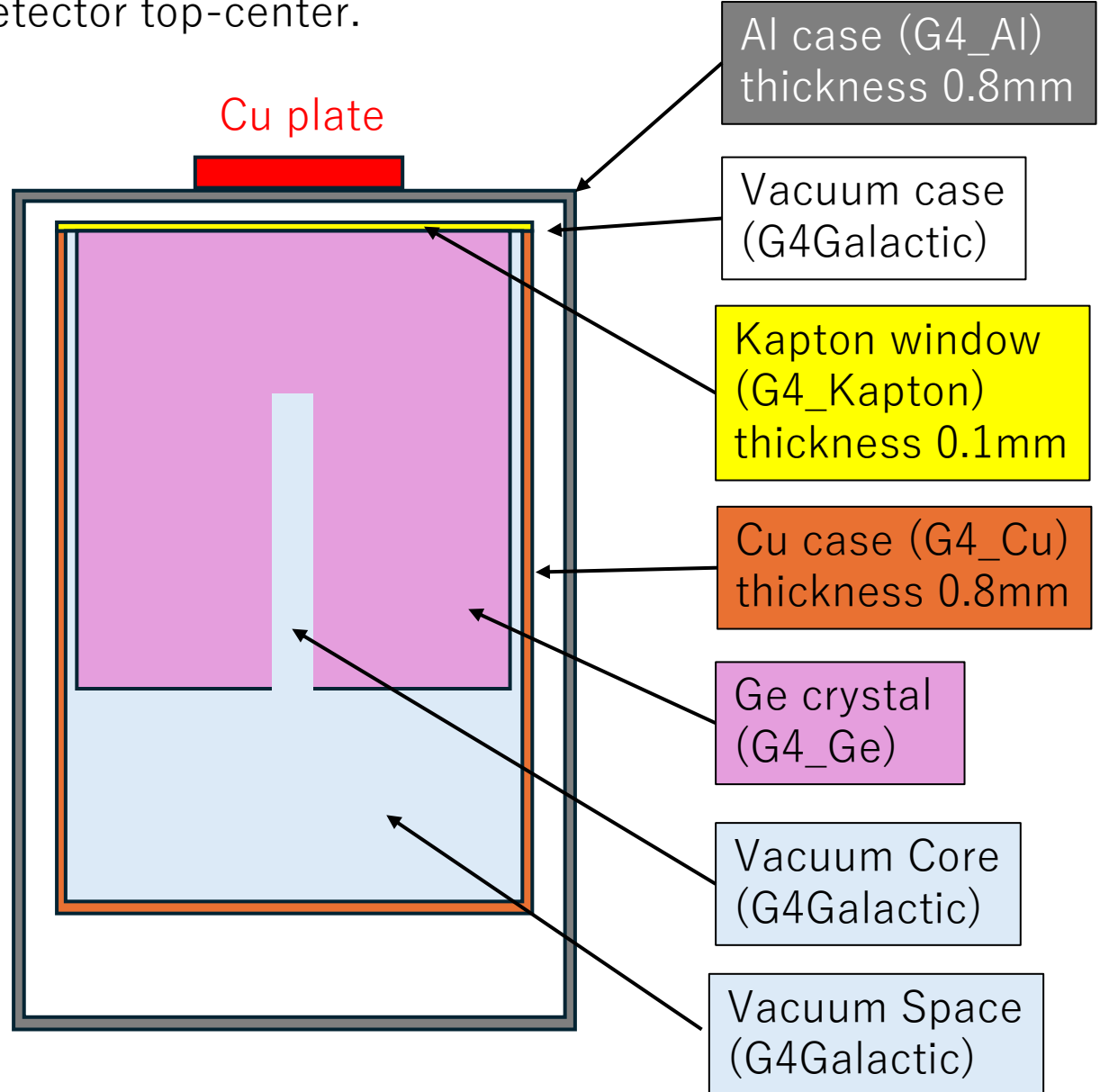
Energy vs Detection efficiency

Add comments

6. Extra

6.1 Set copper square plate xwidth=20mm, ywidth=20mm, zwidth=5mm on the Ge detector top-center.

Check overlaps.



6.2 Add record the mean energy deposition point of each event into root/csv file with tuple format .

Tuple formant

- Name : Edep
- 0th column : Event number (integer)
- 1st column : Total energy deposit in Ge crystal (double)
- 2nd column : Primary vertex position x [mm] (double)
- 3rd column : Primary vertex position y [mm] (double)
- 4th column : Primary vertex position z [mm] (double)
- 5th column : Primary particle direction vector x (double)
- 6th column : Primary particle direction vector y (double)
- 7th column : Primary particle direction vector z (double)
- 8th column : Mean energy deposition point x [mm] (double)
- 9th column : Mean energy deposition point y [mm] (double)
- 10th column : Mean energy deposition point z [mm] (double)

$$\vec{x}_{mean} = \frac{\Sigma(\vec{x}_i \cdot \varepsilon_i)}{\Sigma \varepsilon_i}$$

$$\vec{x}_i = (\overrightarrow{PrestepPoint} + \overrightarrow{PoststepPoint})/2$$

ε_i : Energy deposit

Hint: Check $\Sigma \varepsilon_i = 0$ case and fill dummy values at that case.

6.3 Generate 1000000 event of 1.0 MeV gamma inside the copper plate uniformly and uniformed direction . Make a histogram of energy deposition in Ge crystal and calculate the efficiency of detecting full energy deposition (gamma events shown as 1.0 MeV). Make figure of primary vertex positions and mean energy deposit positions.

[Past your macro file]

[Figure of Total energy deposit in Ge crystal]

Detection efficiency of 1 MeV gamma event: _____ $\pm$ _____ [stat.]

[Figure of Primary Positions]

[Figure of Mean energy deposit Positions]

6.4 Generate 1000000 event of 1.0 MeV gamma from the copper plate surface uniformly and uniformed direction . Make a histogram of energy deposition in Ge crystal and calculate the efficiency of detecting full energy deposition (gamma events shown as 1.0 MeV). Make figure of primary vertex positions and mean energy deposit positions

[Past your macro file]

$$\Sigma \varepsilon_i$$

[Figure of Total energy deposit in Ge crystal]

Detection efficiency of 1 MeV gamma event: _____ $\pm$ _____ [stat.]

[Figure of Primary Positions]

[Figure of Mean energy deposit Positions]

6.5 Describe the difference of observed energy spectra between from inside the material and surface. Describe how these spectra can be distinguished?

Appendix. Precise Ge detector

